

The impact of reservoir scale on amplitude variation with offset

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ABSTRACT

To investigate the vertical and horizontal impact of reservoir scale on the amplitude-versus-offset characteristics, we conduct seismic numerical simulations on models containing spatially confined lithologic units with different scales. We find that the reservoir scale has a nonlinear effect on the amplitude-versus-offset intercepts and gradients. As the reservoir width increases, amplitude-versus-offset intercept and gradient both first increase, then decrease, and finally remain stable. The amplitude-versus-offset intercept is maximum when the reservoir width is 80% larger than the Fresnel zone radius, whereas the amplitude-versus-offset gradient peaks at 1.5 times the Fresnel zone radius. Both amplitude-versus-offset intercept and gradient are approximately proportional to the reservoir width prior to reaching their maxima. When the lateral extent of the reservoir is more than three times the Fresnel zone radius, the amplitude-versus-offset attributes are constant. Modelling the reservoir thickness shows that intercept and gradient behave in a manner similar to that of tuning of thin beds. Both the amplitude-versus-offset intercept and gradient first increase and then decrease with the thickness, peaking at the tuning thickness. The thickness contribution to amplitude-versus-offset variations is negligible when the thickness is larger than 1.6 times of the tuning thickness. Considering the magnitude of the changes in amplitude-versus-offset intercept and gradient caused by reservoir scale, the width causes a maximum 433% intercept increase and a 344% gradient increase, whereas the thickness causes a maximum 100% intercept increase and a 73% gradient increase. Cross-plotting the amplitude-versus-offset intercept and gradient shows the reservoir scale change gives rise to an anti-clockwise spiraling effect. In conclusion, the lateral and vertical extents of the reservoir both play an important role in amplitude variation with offset. Our analysis shows that the lateral reservoir extent has a larger impact on the amplitude variation with offset than the vertical tuning effect.

Key words: AVO, Seismic amplitude, Resolution.

1 INTRODUCTION

Since Ostrander (1984) pointed out that amplitude increased with offset in gas sands, amplitude variation with offset or angle (AVO or AVA) has been studied by many geophysicists and is widely applied to the discrimination of lithology

and fluid (Smith and Gidlow 1987; Rutherford and Williams 1989; Fatti *et al.* 1994; Castagna and Swan 1997; Foster, Keys, and Lane 2010). AVO has also been extensively applied to prospects with finite reservoir scale, such as the karst reservoir in the Tarim basin in China, which is famous for its numerous cavernous fractured reservoirs (Zeng *et al.* 2011; Yang *et al.* 2012; Sun *et al.* 2012; Zhao *et al.* 2014). Traditional AVO analysis is based on the Zoeppritz equations

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(Zoeppritz 1919) whose derivation is based on the assumption of a single interface between two elastic half-spaces. It is therefore in principle not suitable to the analysis of pre-stack amplitudes and AVO responses in finite-scale settings. However, in our opinion, AVO can be extended to arbitrary interfaces through diffraction theory. Based on Huygens' principle, Trorey (1970) showed that a subsurface reflector can be represented as an ensemble of diffractors, with each diffraction point as the source of a secondary wave. The seismic signal measured at the receivers placed on the surface is thus the sum of all diffraction responses. The single interface, as used by the Zoeppritz equations, is a simple special case of Trorey's diffraction theory. This leads us to conclude that AVO can also be analysed in the case of a diffracting reservoir.

We believe the reservoir scale plays a crucial role in AVO characteristics, particularly for reservoirs at seismic scale. For reservoirs at finite scales, both laterally and vertically, the identification of lithology and fluid through AVO could be contaminated by the effect of the reservoir scale. For quantitative AVO analysis, and in particular the analysis of lithology and fluid content, reservoir scale and resolution, both laterally and vertically, needs to be understood, in order to avoid misinterpretation. A common approximation of the spatial resolution is the Widess resolution L (Widess 1973):

$$L = \lambda/8, \quad (1)$$

where λ is the wavelength. According to Widess, the information on the layer thickness is encoded in the amplitude of the composite reflection. The tuning thickness $\lambda/4$ is twice the spatial resolution. Lateral resolution is usually estimated as the Fresnel zone radius, given by (Sheriff 1977):

$$L_f = \sqrt{\left(b + \frac{\lambda}{4}\right)^2 - b^2}, \quad (2)$$

where b is the target depth and λ is the wavelength. The boundary of the (first) Fresnel zone corresponds to the position of maximum energy build-up (Brühl, Vermeer, and Kiehn 1996). The first Fresnel zone is often used as an indicator of the region that contributes energy (constructively) to the total reflection energy, whereas energy returning from areas outside this region interferes destructively.

For the thickness effect, previous researchers have conducted a large number of AVO tuning studies (Bakke and Ursin 1998; Liu and Schmitt 2003). The AVO response of a thin layer is dependent upon the thickness of the layer and differs significantly from the AVO response of a simple interface (Almoghribi and Lange 1986; Juhlin and Young 1993). Swan (1988) pointed out that AVO distortions due to tuning might

be larger than the underlying lithologic AVO effect. Vertical tuning analysis is closely related to Widess's analysis (1973). However, we should not analyse the thickness impact without considering the width impact. Actually, the AVO of a reservoir is the compositive result of both thickness and width impacts. Few well-developed conclusions about the effect of the reservoir width on AVO have been published. Wu *et al.* (2006) demonstrated significant effects of wave diffractions due to lateral changes in the overburden properties on AVO. Wang *et al.* (2002) suggested to decompose the structural amplitude effect (focusing and defocusing) from the AVO attributes. The study conducted by Sun *et al.* (2012) showed the AVO responses of karst caves with different diameters and suggested AVO intercept and gradient were affected by the cave size. Xu, Di, and Wei (2014) pointed out that both reservoir thickness and width exerted considerable effects on AVO. However, the reservoir scale in their study is so limited to be systematic enough, and it is mostly less than one wavelength in particular.

In this paper, our objective is to demonstrate the impact on AVO of reservoir scale, by which we mean the combined effect of reservoir width and thickness. Rather than generating AVO angle gathers using the Zoeppritz equations, we conduct finite difference modelling of the elastic seismic wave equation. We model reservoirs over a large range of scales allowing us to map out the lateral and vertical tuning effects of AVO intercept and gradients. We will show that the reservoir scale has a considerable impact on the AVO behaviour.

2 SEISMIC NUMERICAL SIMULATION

We performed 2D numerical simulations of the elastic wave equation with staggered-grid finite-difference method on models including reservoirs with extensive scale ranges. Since our goal was to demonstrate the impact of reservoir scale on amplitude variation with offset (AVO), all parameters other than the scales of the reservoir were held fixed in all numerical simulations. A Ricker wavelet was employed with the dominant frequency of 30 Hz. The calculating grid was 10 m. For each model, 220 shots gathers were calculated, and there were 120 receivers within each shot, meaning the maximum offset was 2,400 m. The receiver interval was 20 m, whereas the source interval was 40 m. There were 3,600 sampling points within each trace and the sampling interval was 0.5 ms.

To investigate the impact of the reservoir width on AVO, we first designed two three-layer models containing reservoirs with a wide range of widths (100 m, 200 m, 300 m, 400 m, 500 m, 600 m, 700 m, 800 m, 900 m, 1,000 m, 1,100 m, and 1,200 m) within the second layers as shown in Fig. 1(a, b).

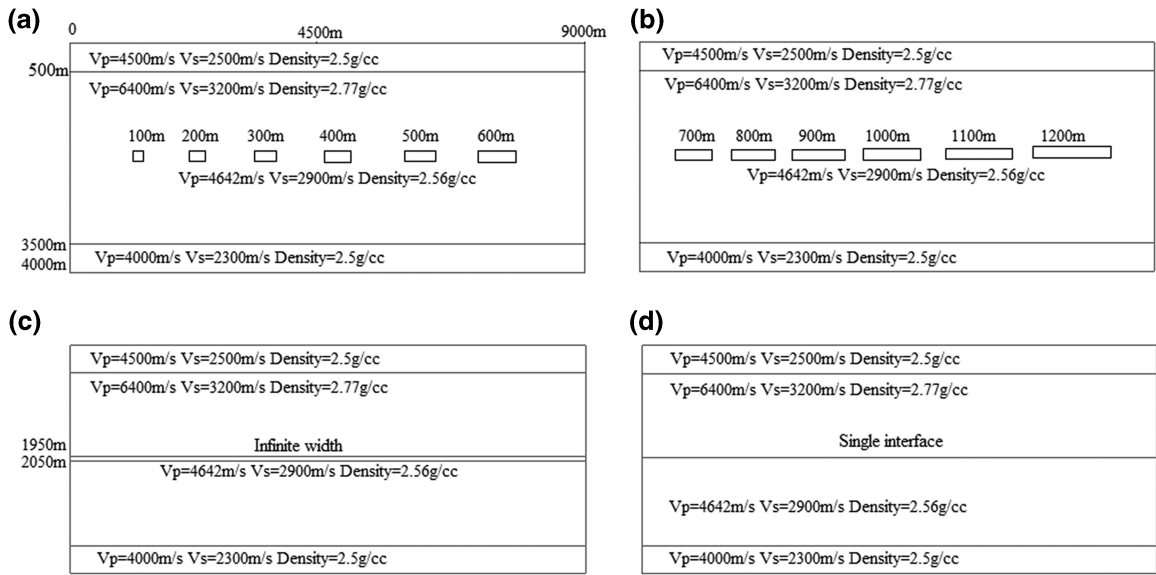


Figure 1 Three-layer models containing reservoirs with different widths: (a) ranging from 100 m to 600 m, (b) ranging from 700 m to 1,200 m, (c) infinite width, and (d) single interface.

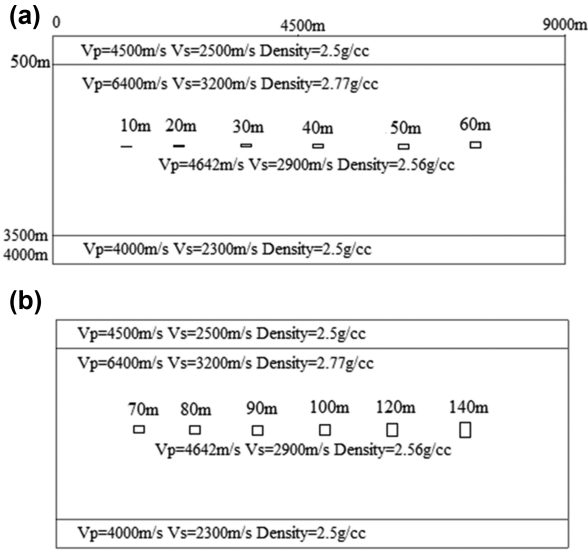


Figure 2 Three-layer models containing reservoirs with different thicknesses: (a) ranging from 10 m to 60 m and (b) from 70 m to 140 m.

Considering the extreme situation, we designed a model with an infinite-width reservoir similarly located at the middle of the second layer, shown in Fig. 1(c). The reservoir width is the same as the layers (9,000 m). The reservoir thickness in these models is held constant at 100 m. In addition, we also designed a model of a single interface at the middle of the second layer, as given in Fig. 1(d). The elastic parameters of the top and base media of the single interface are the

same as that of the surroundings and reservoirs in the above models.

Similarly, to demonstrate the impact of the reservoir thickness on AVO, we designed two three-layer models containing reservoirs with a wide range of thicknesses (10 m, 20 m, 30 m, 40 m, 50 m, 60 m, 70 m, 80 m, 90 m, 100 m, 120 m, and 140 m) within the second layers, as shown in Fig. 2. For these models, the lateral reservoir extent is held constant at 100 m. The parameters within our models originate from the real karstic reservoir strata from Tarim Basin, China. It is deeply buried, more than 5,500 m, and the velocity and density are very high (Zhao *et al.* 2014; Xu *et al.* 2016). Field data from the area show that the P-wave velocity is more than 6,000 m/s.

3 MODELLING RESULTS

3.1 Modelling gathers in offset and angle domain

Based on the raw seismic data, we obtained from seismic simulations with the above parameters, we conducted a conventional processing procedure including amplitude compensation, noise filtering, removing multiples, deconvolution, normal movement correction, and stack. Figure 3(a) is an example of one of the processed shot gathers. The waveforms of the reservoir top and base interfere with each other and appear between 680 ms and 800 ms. To get the amplitude-versus-offset (AVO) variation in the angle domain, we transformed the shot gather from the offset domain to the angle domain,

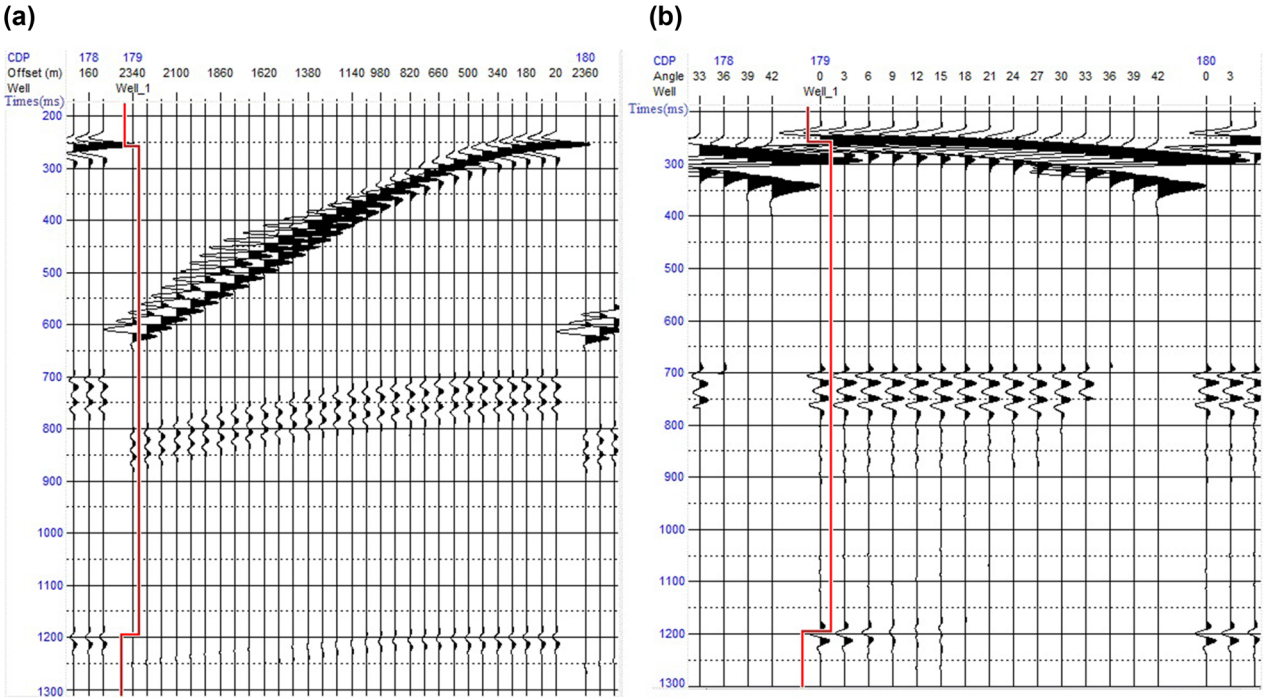


Figure 3 Shot gather in (a) offset domain and (b) angle domain for the midpoint of the 400-m-wide reservoir, corresponding to CMP 179, within the model in Fig. 1(a). The waveforms of the reservoir top and base interfere with each other and appear from about 680 ms to 800 ms.

as shown in Fig. 3(b). The transform process needs a well, including information of interfaces and elastic parameters. We used an artificial well that was generated based on Fig. 1(a). The angle transform process is based on the ray parameter equation reported by Bale, Leaney, and Dumitru (2001):

$$\theta = \sin^{-1} \left((x V_{int}) / (V_{rms}^2 t) \right), \quad (3)$$

where θ is the incidence angle, x is the offset, V_{int} is the interval velocity of one layer, V_{rms} is the root mean square (RMS) velocity, and t is the two-way time at offset x . From Fig. 3(b), we can see an obvious waveform truncation for far angles after 33° . That is because the angle of incidence for a constant offset trace decreases with the depth. The maximum incidence for the reservoirs near 2,000 m deep, which is about 33° , is less than that for the above interface, which is more than 42° . In addition, the incidence range for the 3,000 m interface is much narrower.

Similar to Trorey's (1970) work, we illustrate in Fig. 4a zero offset section near the 400-m-wide reservoir in Fig. 1(a) to show the diffractions. The reservoir ranges from CDP 160 to CDP 199. As mentioned above, the artificial well, Well_1, is located at the middle of the reservoir (CDP 179). As reported by Sun and Bancroft (2003), when the reservoir width is large enough, we can get a good image of the reservoir except at

the reservoir edges, where the diffractions are not properly collapsed. We can see uncollapsed diffractions at the left side of the section. In agreement with Trorey's report, we find that the diffraction amplitude is much less than the reflections of the reflector. The events at the right of CDP 240 are responses for the 500-m-wide reservoir. It is hard to find diffractions between the two reservoirs. That is because the two reservoirs are so close that the diffractions destructively interfere with each other.

3.2 AVO extraction

In our experiment, the wavelength computed using the velocity of the reservoirs is estimated to be about 154 m. The spatial resolution is calculated as about 19 m, and the lateral resolution is calculated as about 394 m through equations (1) and (2). In the above models, all reservoir thicknesses, ranging from 10 m to 140 m, are comparable to the spatial (vertical) resolution. Similar to a thin bed, the reflections of the top and base of a finite-scale reservoir interfere with each other. With different reservoir scales, the interference between the top and base of reservoirs is different. Thus, the apparent amplitude of a finite-scale reservoir is controlled by the top and base reflection coefficients and the reservoir scale. Different from

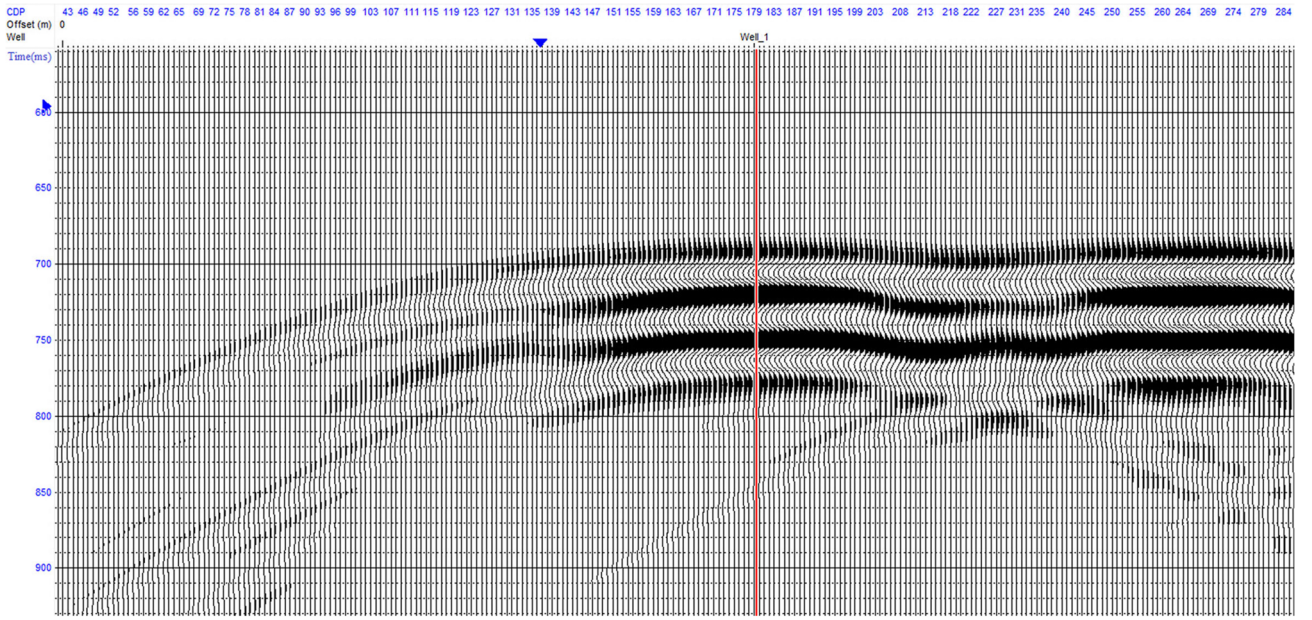


Figure 4 A part of the zero-offset section for the model in Fig. 1(a).

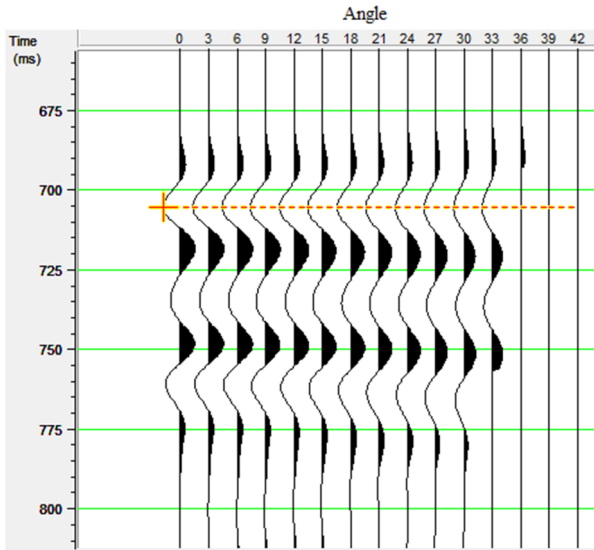


Figure 5 Waveforms for the 400-m-wide reservoir from Fig. 1(a). The first trough around 705 ms, as marked with a dashed line, in each trace is extracted as the amplitude during AVO extraction.

conventionally extracted reflection amplitudes of the top interface during AVO analysis, we can only get the composite reflections of the top and base for a finite-scale reservoir.

From the angle gathers at the common midpoints (CMPs) of reservoirs, we extract the amplitudes of different incident angles to investigate the impact of width and thickness on AVO. Figure 5 illustrates the amplitude extraction process.

The first troughs of the composite waveform of the reservoirs are extracted for AVO analysis. Least-squares fitting is used to extract AVO intercepts and gradients based on Shuey's two-term approximation of the Zoeppritz equations (Shuey 1985):

$$R(\theta) = A + B \sin^2 \theta, \quad (4)$$

where θ is the incidence angle, A is the AVO intercept, and B is the AVO gradient. Then, we can demonstrate the impact of the width and thickness on AVO characteristics.

3.3 Impact of reservoir width on AVO

AVO curves of reservoirs with different widths are shown in Fig. 6. AVOs of these reservoirs belong to class IV. As Castagna, Swan, and Foster (1998) explained, class IV AVO frequently occurs when a low-velocity sand is overlain by a high-velocity unit. This coincides with the situation in our models. We can see that AVO dramatically varies with the reservoir width. To be more precise, the magnitude of the amplitude first increases with the width when the width ranges from 100 m to 700 m, and then decreases with the width when the width ranges from 800 m to 1,200 m. Noticeably, the ordinate ranges of Figs. 6(a) and (b) are different. In fact, the amplitude decrease in Fig. 6(b) is very slight in comparison with the amplitude increase in Fig. 6(a). In Fig. 6(b), the AVO curve of the 1,200 m almost overlaps with that of the infinite width. Eventually, the AVO curve tends to the infinite-width

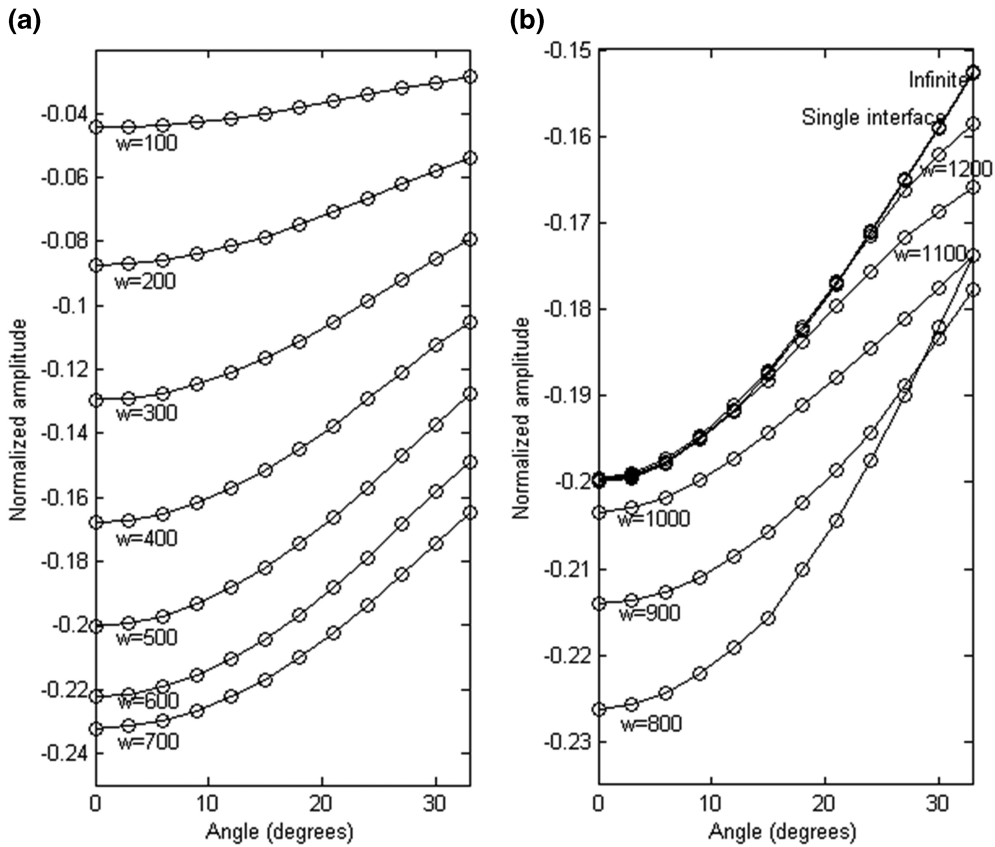


Figure 6 AVO curves of reservoirs with different widths: (a) ranging from 100 m to 700 m, and (b) from 800 m to 1,200 m. Empty circles indicate amplitude samples from AVO extraction.

situation. The magnitude of amplitudes peaks at 700 m, which is about 1.8 times of the Fresnel zone radius.

To show the width impact more clearly, we extract the intercepts (*A*) and gradients (*B*) of the AVO curves in Fig. 6, given in Table 1, and illustrate them in Fig. 7. Intercept and gradient clearly both depend on the reservoir width, ranging from 100 m to 1,200 m. Given that the intercept measures the amplitude at normal incidence, the intercept variation is in accordance with the amplitude variation discussed in the last paragraph. The magnitude of the intercept shows an approximately linear increase from 0.044 at 100 m to 0.233 at 700 m, which is an increase of about 430%. Then, it is followed by a slight drop until 1,000 m, which is almost 2.5 times of the Fresnel zone radius. Beyond a reservoir width of 1,000 m, the intercept is constant. By comparison, the gradient behaves quite differently. In particular, it varies beyond the 1,000-m reservoir width, at which the intercept becomes stable. The gradient shows a slight rise from 0.12 at 1,000 m to 0.16 at 1,200 m. When it is less than 1,000 m, the magnitude of the gradient also shows an approximately linear increase from

Table 1 AVO intercepts (*A*) and gradients (*B*) of reservoirs with different widths and the same thickness of 100 m from Fig. 6

width (m)	<i>A</i>	<i>B</i>
100	−0.044	0.057
200	−0.087	0.120
300	−0.129	0.175
400	−0.167	0.224
500	−0.200	0.252
600	−0.222	0.258
700	−0.233	0.233
800	−0.227	0.177
900	−0.214	0.129
1000	−0.203	0.116
1100	−0.198	0.134
1200	−0.198	0.158

0.057 at 100 m to 0.258 at 600 m, which is an increase of about 353%. Then, it is followed by a decrease until 1,000 m (0.116). The gradient peaks at 6,000 m, which is about 1.5 times of the Fresnel zone radius.

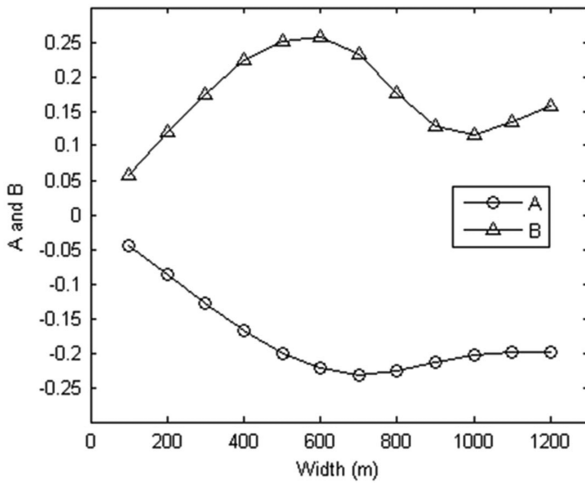


Figure 7 AVO intercept (A) and gradient (B) versus reservoir width from AVO curves in Fig. 6.

3.4 Impact of reservoir thickness on AVO

AVOs for different thicknesses are shown in Fig. 8. From Fig. 8(a), it is easy to see AVO curves of different thicknesses

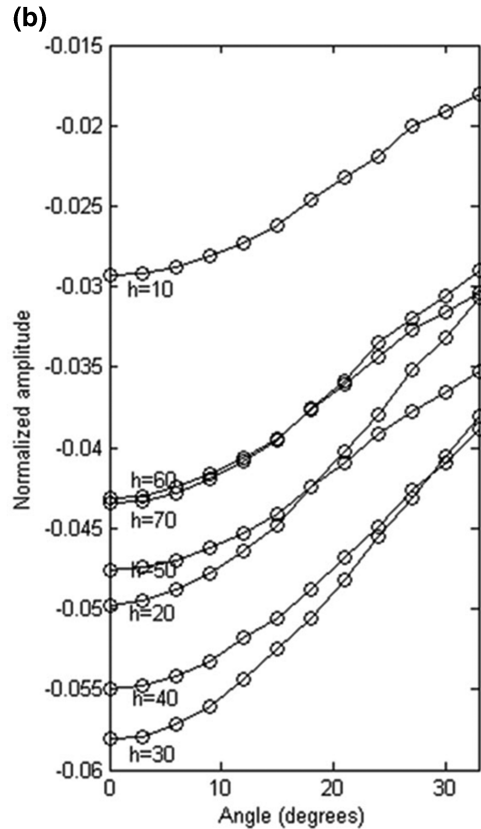
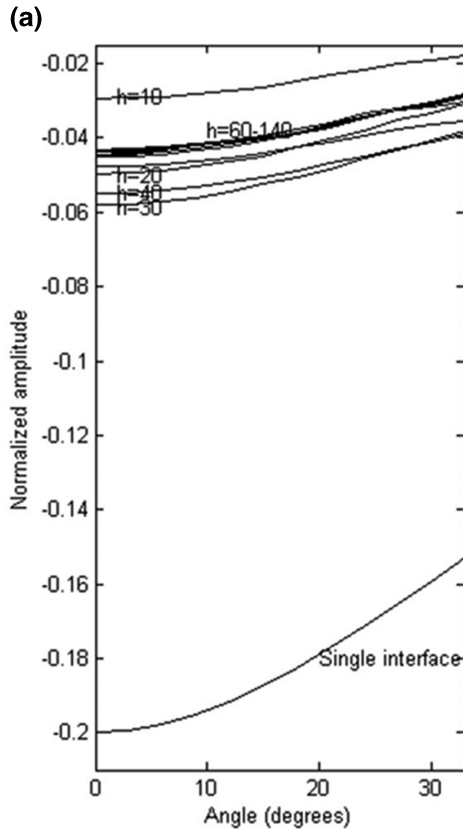


Figure 8 AVO curves of reservoirs with different thicknesses (a) ranging from 10 m to 140 m. AVO curves from 10 m to 70 m are magnified in (b). Empty circles indicate amplitude samples from AVO extraction.

dramatically deviate from that of the single interface. That is because the widths of the former are only 100 m, whereas that of the latter is infinite (large compared with the seismic wavelength). As discussed above, the amplitudes generally increase with the reservoir width. Therefore, the amplitudes of the single interface are much larger than that of the reservoirs in Fig. 1(c). We can also see the variation of AVO curves with the thickness is much less pronounced than that with the width. AVO curves corresponding to thicknesses ranging from 60 m to 140 m almost overlap with each other. That means the thickness scarcely contributes to the AVO variation when the thickness is more than 60 m. To see the variation with thicknesses from 10 m to 70 m more clearly, we magnify their AVO curves in Fig. 8(b). The magnitude of the amplitudes first increases and then decreases with the thickness, peaking at the thickness of 30 m. The curve of the 60 m almost overlaps with that of the 70 m. The thickness change will not cause obvious AVO changes when it is more than 60 m.

Table 2 gives the extracted the intercepts (A) and gradients (B) of the AVO curves in Fig. 8(a), and we illustrate them in Fig. 9 to show the thickness impact more clearly. There is

Table 2 AVO intercepts (A) and gradients (B) of reservoirs with different thicknesses and the same width of 100 m from Fig. 8(a)

h/m	A	B
10	-0.029	0.041
20	-0.049	0.066
30	-0.058	0.071
40	-0.054	0.055
50	-0.047	0.044
60	-0.042	0.045
70	-0.043	0.055
80	-0.044	0.057
90	-0.043	0.052
100	-0.043	0.052
120	-0.044	0.053
140	-0.043	0.052

a rise in the magnitude of the intercept from 0.029 at 10 m to 0.058 at 30 m, which is an increase of about 100%. Then, the magnitude of the intercept gradually decreases to 0.42 at 60 m. The magnitude of the intercept remains at this level when the thickness ranges from 60 m to 140 m. The variation of the gradient is similar to that of the intercept, except that there is some fluctuation between 60 m to 90 m. However, the overall trend is almost the same. The gradient peaks at 30 m (0.071), which is close to the tuning thickness, and remains almost stable when the thickness is larger than 60 m, which is 1.6 times of the tuning thickness. The maximum change of the gradient due to the thickness is about 73% (from 0.041 at 10 m to 0.071 at 30 m). The impact of reservoir thick-

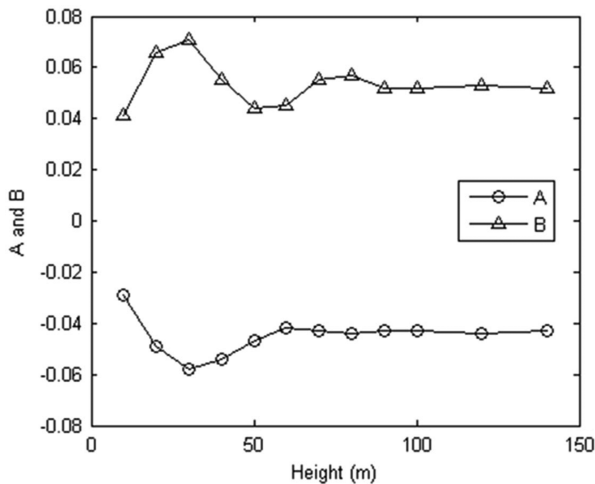


Figure 9 AVO intercept (A) and gradient (B) versus reservoir thickness from AVO curves in Fig. 8.

ness could be explained by Widess theory (Widess 1973). As pointed out by Chopra, Castagna, and Xu (2009), in Widess' theory, the reflection coefficients of the thin bed top and bottom are exactly equal and opposite. In our study, the above and laying backgrounds of reservoirs are the same medium. Therefore, the Widess limit applies in our situation.

4 DATA INTERPRETATION

As mentioned above, the amplitude-versus-offset (AVO) intercepts, the amplitudes at zero offset, first increase with the width (from 100 m to 700 m), followed with a slight decrease (from 700 m to 1,000 m), and finally remain stable. The process is similar to the tuning effect for a thin bed. We conjecture that there is a tuning effect in the lateral direction. The tuning width is 1.8 times of the Fresnel zone radius, which is 700 m here. The waveform energy increases with the width before the tuning width and decreases after the tuning width. When the width is larger than 3 times of the Fresnel zone, the waveform energy will remain stable.

The AVO curve of a reservoir 1,200 m wide almost overlaps with that of the infinite width as shown in Fig. 6(b). The largest width that contributes to AVO change, which is 1,200 m here, is about three times of the Fresnel zone radius. That is because the reservoir width would not contribute to seismic energy change when it is beyond the lateral resolution. Consequently, as the width increases, the AVO curve tends to the infinite-width situation. The AVO curves of the reservoir with infinite width and of a single interface are clearly identical. These reservoirs differ only in thickness, the former being 100 m and the latter infinitely thick. However, as discussed, AVO attributes are not affected by vertical tuning beyond a thickness of 60 m (using our model parameters).

It is worth noting that reservoir scale can cause considerable variations on cross-plots of AVO intercepts and gradients. Cross-plotting AVO intercepts and gradients is a common approach during AVO analysis. As pointed out by Castagna *et al.* (1998), gas sands tend to plot to the lower left of the brine sands on cross-plots of A and B, meaning A and B of gas sands are more negative than that of brine sands. The deviation of gas sands from brine sands is the basis for many hydrocarbon detection methodologies. However, we see here that the deviation may also be caused by non-petrophysical factors such as the reservoir scale. This is illustrated in Fig. 10, which cross-plots AVO intercepts and gradients of different widths and thicknesses for reservoirs in the above models. Since AVO in this study belongs to class IV, samples fall in quadrant II. The reservoir scale change, both

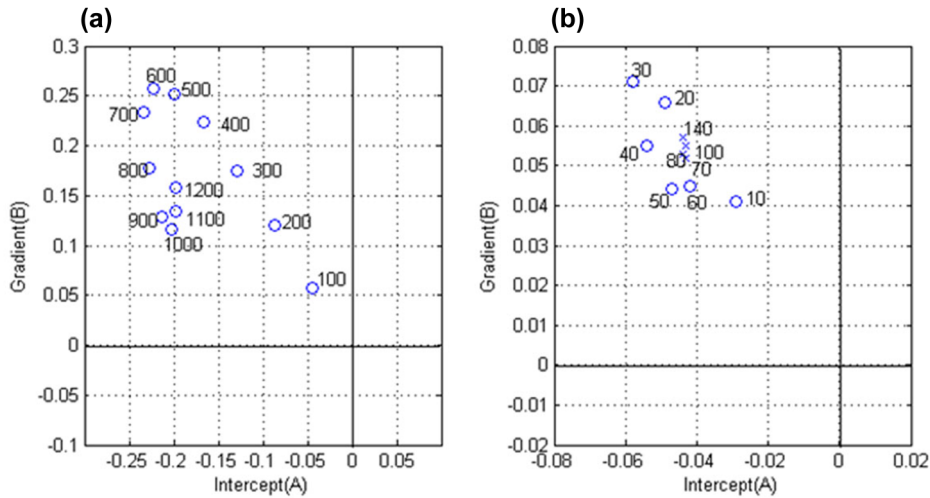


Figure 10 Cross-plots of AVO intercepts and gradients of (a) different widths ranging from 100 m to 1,200 m and (b) different thicknesses ranging from 10 m to 140 m. Note that numbers are widths and thicknesses in metres.

the width and thickness, gives rise to a spiraling effect on the intercept-gradient cross-plot. Samples in Fig. 10(a, b) both move in an anti-clockwise direction along with the increase in reservoir scale. This phenomenon coincides with Simm and Bacon's report (Simm and Bacon 2014). In this paper, we studied class IV AVO, whereas in Simm and Bacon's (2014) report, they studied class III AVO. In both situations, points in the intercept and gradient cross-plot rotate in the anti-clockwise direction. This phenomenon reflects the impact of reservoir scale, which is a non-petrophysical factor, no matter which quadrant points fall in. It is reasonable to conjecture that this phenomenon applies for other AVO classes (I and II). Moreover, the magnitude of the deviation caused by width is larger than that caused by thickness. Noticeably, the deviation caused by reservoir scale is comparable to, sometimes even larger, than that caused by hydrocarbons, which was reported by Castagna *et al.* (1998). Therefore, it is reasonable to believe the validity of hydrocarbon detection without considering the impact of reservoir scale is questionable. The reservoir scale is an important factor to be taken into consideration during AVO analysis for reservoirs at finite scale.

5 DISCUSSION

We have presented an analysis of the impact of lateral and vertical scales on the amplitude-versus-offset (AVO) intercept and gradient. This is a single-factor analysis, in the sense that only the size of the reservoir was varied, and all modelling parameters were held constant throughout. Clearly, there are other factors that need to be analysed, and we hope to do this

in the future. This includes the frequency bandwidth, sampling grid, noise, and acquisition density during the modelling process.

Trorey (1970) showed that, at a diffracting edge, the amplitude of the reflection is halved. Similarly, Sun and Bancroft (2003) pointed out that the edge impact on AVO was significant if the reflector was comparable to the Fresnel zone. However, for a large reflector, the diffraction effect is insignificant, particularly in the middle of the reflector that is the furthest point from the edge. As Fig. 4 shows, when the reservoir width is large enough, we can get a good image of the reservoir, except there is serious diffraction near the edge. Therefore, to avoid the edge diffraction unreliability, all the above AVO analysis is conducted on seismic gathers for the middle point of each reservoir. We can extract AVO for reservoirs with higher reliability from the properly collapsed diffractions far from the reservoir edges.

Our conclusions are drawn from a numerical model and have yet to be verified with experimental data. Xu *et al.* (2014) conducted physical modelling experiments to understand the reservoir scale impact. However, the range of reservoir size analysed is too small for us to be able to verify if their results match ours.

The seismic response of spatially confined units is mainly dominated by wave diffraction, instead of simple reflections at a single interface. Diffractions will no longer interfere when the unit is large enough. Compared with previous research, we extend the reservoir scale to a wider range and get a more extensive understanding of the impact of reservoir scale on AVO. It is noticeable that there are both disadvantages and

advantages of the impact of the reservoir scale on AVO. On one hand, it brings about contamination to fluid discrimination. On the other hand, the relationship between the reservoir scale and AVO responses may be useful in quantitative interpretation of reservoirs. We suggest that we could make the best of it to achieve reserve estimation. The impact of reservoir scale on AVO we propose in the paper is essential for researchers of AVO technology.

6 CONCLUSIONS

For reservoirs with finite scale, the reservoir scale plays an important role in amplitude variation with offset (AVO) and has a nonlinear impact on AVO intercept and gradient. AVO intercept and gradient are linearly proportional to the reservoir scale only when it is less than some certain values. Based on the data in our numerical simulations, we find AVO intercept achieves its maxima at a width of 1.8 times of the Fresnel zone radius, whereas AVO gradient peaks at a width of 1.5 times of the Fresnel zone radius. The AVO did not change with the width if it is larger than three times of the Fresnel zone radius. Both the AVO intercept and gradient first increase and then decrease with the thickness, peaking at the tuning thickness. The thickness contribution to AVO variations is negligible when the thickness is larger than 1.6 times of the tuning thickness. Compared with the reservoir thickness, the reservoir width is much more dominant in contributing to AVO. On cross-plots of AVO intercept and gradient, the reservoir scale change causes a spiraling effect. Samples move in an anti-clockwise direction along with the increase in reservoir scale.

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